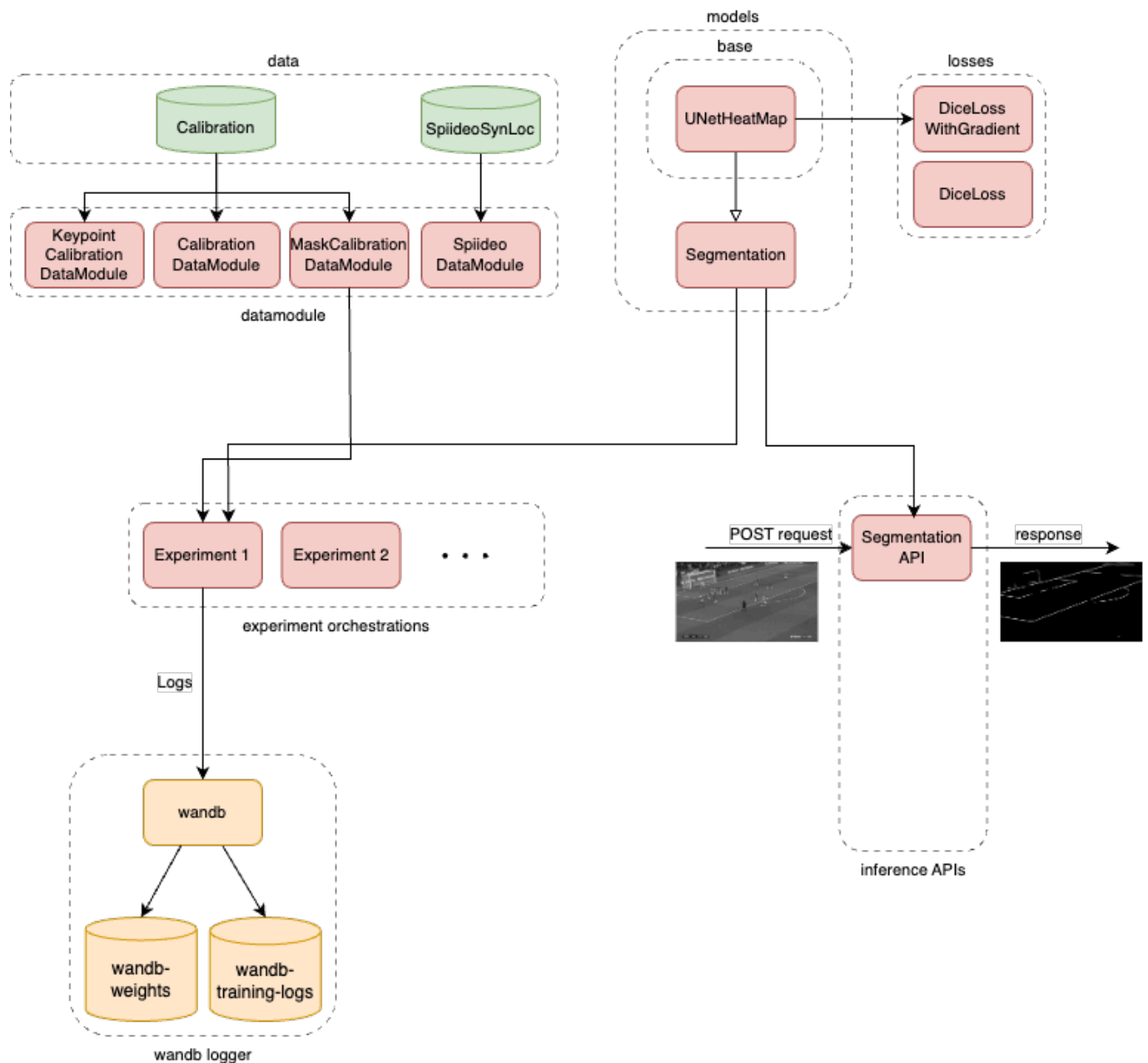


ISC 2, Checkpoint 1

Infrastructure



- ⚠ Wandb account is currently on Trial and will expire soon.
- Considerations to apply for Wandb for Academics.

Task Description

- <https://github.com/Spiideo/sskit>
- <https://www.scitepress.org/publishedPapers/2025/131082/pdf/index.html>
- Dataset includes 40k+ training / $\sim 7k$ validation/ $\sim 9k$ test images of football matches, synthetically generated from 3D rendering techniques.
 - COCO format: image, annotations, categories.
- Features per image:
 - Calibration parameters:
 - Camera matrix: $P_{3 \times 4} = [R_{3 \times 3} \mid \tilde{t}]$
 - Industrial distortion model:
 - Normalised polar coordinates:
 - $(u, v) = \frac{1}{w} \left(x - \frac{w}{2}, y - \frac{h}{2} \right)$
 - $r = \sqrt{u^2 + v^2}$
 - Distortion: $r_{\text{dist}} = p_{\text{dist}}(\tan^{-1}(r_{\text{undist}}))$
 - Undistortion: $r_{\text{undist}} = \tan(p_{\text{undist}}(r_{\text{dist}}))$
 - p_{dist} and p_{undist} are given polynomials. $p_{\text{dist}}^{-1} \approx p_{\text{undist}}$
- Features per each annotated player:
 - Keypoints
 - 2 keypoints recorded, at the pelvis and at its projection to the ground.
 - Both the world coordinates (3D, in world meters) and the image coordinates (2D, in image pixels) are provided.
 - **Location on the pitch** (in meters)
 - Bounding box
- Task: **determine the pitch location of every player.**



Figure 1: Spiideo synthetic dataset. Each player is annotated by two keypoints: the pelvis and the projection of the pelvis onto the field ground.



Figure 2: Spiideo synthetic dataset, after undistortion.

- Metric: *mAP-LocSim*

▸ $\text{LocSym}(\mathbf{x}_1, \mathbf{x}_2) = \exp\left(\ln 0.05 \cdot \frac{d(\mathbf{x}_1, \mathbf{x}_2)^2}{\tau^2}\right)$, where $\tau = 1$ m is the tolerance constant $\rightarrow$ “target on precise location”

Ideas

Approach	Drawbacks	Status
• Train a YOLO detector to detect regions associating to each player • Determine the keypoints for each player based on the cropped region	• Images are blurry and contain artifacts $\rightarrow$ harder to detect objects	experiment
• Segmentation model (UNet-based) • Determine the keypoints for each player based on the cropped region	• UNet isn't stable, may produce false negative pixels. This behaviour can be seen from the MaskCalibration model.	in research
• Gaussian Mixture Model (?) • Treat each player as a Gaussian distribution (heatmap, etc.)	-/-	in research